whether Maxwell's theory allows a particle with one unit of electric charge to disintegrate into particles with two units of electric charge, this is a qualified no.

The fact is, no one has shown that the laws of physics absolutely rule out past-directed time travel. To the contrary, some physicists have even laid out hypothetical instructions for how a civilization with unlimited technological prowess, operating fully within the known laws of physics, might go about building a time machine (when we speak of time machines, we will always mean something that is able to travel both to the future and to the past). The proposals bear no resemblance to the spinning gizmo described by H. G. Wells or Doc Brown's souped-up DeLorean. And the design elements all brush right up against the limits of known physics, leading many researchers to suspect that with subsequent refinements in our grasp of nature's laws, existing and future proposals for time machines will be deemed beyond the bounds of what's physically possible. But as of today, this suspicion is based on gut feeling and circumstantial evidence, not solid proof.

Einstein himself, during the decade of intense research leading to the publication of his general theory of relativity, pondered the question of travel to the past.[10] Frankly, it would have been strange if he hadn't. As his radical reworkings of space and time discarded long-accepted dogma, an ever-present question was how far the upheaval would go. Which features, if any, of familiar, everyday, intuitive time would survive? Einstein never wrote much on the issue of time travel because, by his own account, he never made much progress. But in the decades following the release of his paper on general relativity, slowly but surely, other physicists did.

Among the earliest general relativity papers with relevance for time machines were those written in 1937 by the Scottish physicist W. J. van Stockum[11] and in 1949 by a colleague of Einstein's at the Institute for Advanced Study, Kurt Gödel. Van Stockum studied a hypothetical problem in general relativity in which a very dense and infinitely long cylinder is set into spinning motion about its (infinitely) long axis. Although an infinite cylinder is physically unrealistic, van Stockum's analysis led to an interesting revelation. As we saw in Chapter 14, massive spinning objects drag space into a whirlpool-like swirl. In this